

PC Game Deals - Part A Formula Sheet

This sheet summarises the calculations and logical checks used in the PC Game Deals assessment.

1. Check Whether a Game Is On Sale

A game is on sale when its sale price is lower than its normal price.

$$\text{Sale Price} < \text{Normal Price}$$

Example: Hollow Knight has a normal price of R14.99 and a sale price of R7.49.
 $7.49 < 14.99$, so the game is ON SALE.

If Sale Price = Normal Price, the game is not on sale.

2. Amount Saved

Before calculating the discount percentage, calculate how much money was saved.

$$\text{Amount Saved} = \text{Normal Price} - \text{Sale Price}$$

Example: If a game normally costs R80 and is now R60:
Amount Saved = $80 - 60 = R20$

3. Discount Percentage

The discount percentage shows what percentage of the original price has been removed.

$$\text{Discount \%} = ((\text{Normal Price} - \text{Sale Price}) / \text{Normal Price}) \times 100$$

Example 1: Normal Price = R80, Sale Price = R60
Discount = $((80 - 60) / 80) \times 100$
Discount = $(20 / 80) \times 100$
Discount = 25%

Example 2 - Hollow Knight:
Discount = $((14.99 - 7.49) / 14.99) \times 100$
Discount is approximately 50.03%, which rounds to **50%**.

JavaScript used Math.round() because the assessment asks for a whole-number discount percentage.

4. Current Price

The current price is the price the customer would pay now.

$$\text{If Sale Price} < \text{Normal Price: Current Price} = \text{Sale Price}$$

$$\text{Otherwise: Current Price} = \text{Normal Price}$$

Examples:

Hollow Knight: $R7.49 < R14.99$, so Current Price = R7.49.
Stardew Valley: $R13.99 = R13.99$, so Current Price = R13.99.

5. Finding the Cheapest Game

To find the cheapest game, compare the current price of each game and keep the game with the lowest current price.

Cheapest Game = Game with the lowest Current Price

For this dataset, **Portal 2** is the cheapest game because its current price is **R0.99**.

6. Search Logic

Search is not a mathematical calculation, but it is one of the required Part A operations. Both the game title and the user's search text are converted to lowercase, then the program checks whether the title contains the search text.

lowercase(Game Title) contains lowercase(Search Word)

Example: Search Word = "hol"

Hollow Knight becomes "hollow knight".

"hollow knight" contains "hol", so Hollow Knight is returned.

Quick Reference

Task	Rule / Formula
On sale?	Sale Price < Normal Price
Amount saved	Normal Price - Sale Price
Discount %	$((\text{Normal Price} - \text{Sale Price}) / \text{Normal Price}) \times 100$
Current price	Sale Price if on sale; otherwise Normal Price
Cheapest game	Lowest Current Price
Search	Lowercase title contains lowercase search word